

OS Lab

Practical-5

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
lahari@DESKTOP-AIC33F7:~$ cd ~/OSLab
lahari@DESKTOP-AIC33F7:~/OSLab$ cd OSLab
-bash: cd: OSLab: No such file or directory
lahari@DESKTOP-AIC33F7:~/OSLab$ cd ~/OSLab
lahari@DESKTOP-AIC33F7:~/OSLab$ code producer_consumer.c
lahari@DESKTOP-AIC33F7:~/OSLab$ gcc producer_consumer.c -o producer_consumer
lahari@DESKTOP-AIC33F7:~/OSLab$ ./producer_consumer
Consumer: Received data
First value: 1
Last value: 1000

Producer: Sent 1000 integers
Data transferred: 3.91 KB
Communication time: 0.000998 seconds
```

```

lahari@DESKTOP-AIC33F7:~/OSLab$ ls -l | grep ".c"
-rwxr-xr-x 1 lahari lahari 16272 Aug 29 08:34 cr
-rw-r--r-- 1 lahari lahari 230 Aug 29 08:28 named_createpipe
-rwxr-xr-x 1 lahari lahari 16448 Aug 29 13:08 producer_consumer
-rw-r--r-- 1 lahari lahari 1724 Aug 29 13:07 producer_consumer.c
-rw-r--r-- 1 lahari lahari 268 Aug 29 08:33 section13_read.c
-rw-r--r-- 1 lahari lahari 762 Aug 29 08:31 section13pipescr.c
-rw-r--r-- 1 lahari lahari 310 Aug 29 08:36 section13write.c
-rw-r--r-- 1 lahari lahari 124 Aug 29 09:37 signal.c
-rw-r--r-- 1 lahari lahari 238 Aug 29 09:32 signal1.c
lahari@DESKTOP-AIC33F7:~/OSLab$ cd ~/OSLab
lahari@DESKTOP-AIC33F7:~/OSLab$ code section13write.c
lahari@DESKTOP-AIC33F7:~/OSLab$ gcc section13write.c -o section13write
lahari@DESKTOP-AIC33F7:~/OSLab$ ./section13write
-rwxr-xr-x 1 lahari lahari 16272 Aug 29 08:34 cr
-rw-r--r-- 1 lahari lahari 230 Aug 29 08:28 named_createpipe
-rwxr-xr-x 1 lahari lahari 16448 Aug 29 13:08 producer_consumer
-rw-r--r-- 1 lahari lahari 1724 Aug 29 13:07 producer_consumer.c
-rw-r--r-- 1 lahari lahari 268 Aug 29 08:33 section13_read.c
-rw-r--r-- 1 lahari lahari 762 Aug 29 08:31 section13pipescr.c
-rwxr-xr-x 1 lahari lahari 16312 Aug 29 13:11 section13write
-rw-r--r-- 1 lahari lahari 1223 Aug 29 13:10 section13write.c
-rw-r--r-- 1 lahari lahari 124 Aug 29 09:37 signal.c
-rw-r--r-- 1 lahari lahari 238 Aug 29 09:32 signal1.c

```

```

lahari@DESKTOP-AIC33F7:~/OSLab$ gcc section13write.c -o section13write
lahari@DESKTOP-AIC33F7:~/OSLab$ ./section13write
-rwxr-xr-x 1 lahari lahari 16272 Aug 29 08:34 cr
-rw-r--r-- 1 lahari lahari 230 Aug 29 08:28 named_createpipe
-rwxr-xr-x 1 lahari lahari 16448 Aug 29 13:08 producer_consumer
-rw-r--r-- 1 lahari lahari 1724 Aug 29 13:07 producer_consumer.c
-rw-r--r-- 1 lahari lahari 268 Aug 29 08:33 section13_read.c
-rw-r--r-- 1 lahari lahari 762 Aug 29 08:31 section13pipescr.c
-rwxr-xr-x 1 lahari lahari 16312 Aug 29 13:13 section13write
-rw-r--r-- 1 lahari lahari 1228 Aug 29 13:12 section13write.c
-rw-r--r-- 1 lahari lahari 124 Aug 29 09:37 signal.c
-rw-r--r-- 1 lahari lahari 238 Aug 29 09:32 signal1.c

```